ELEC2134 : Circuits and Signals

PART A (Week 1 to Week 5)

Transform Methods

- Periodic signals and Fourier series
- Aperiodic signals and Fourier transforms
- Laplace transforms of signals and circuits

System Analysis

- Linear systems and convolution
- Impulse response
- Frequency response and transfer functions
- Steady-state analysis of linear systems
- Transient analysis of 1st and 2nd systems
- Circuit analysis using transform methods

PART B (Week 7 to Week 10)

AC Circuit Analysis

- AC circuits with sinusoidal inputs
- Use of phasors and complex impedance in AC circuit analysis
- AC power: real, reactive and apparent power
- Power factor (leading/lagging)
- Series and parallel resonance
- **Transformers and coupled coils**
- Application of network theorems: Two port networks

Adapted by Aous Naman (PhD) from lectures by Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

ELEC2134 : Circuits and Signals (PART B)

Topic 9: Transformers and coupled coils

Magnetically Coupled Circuits

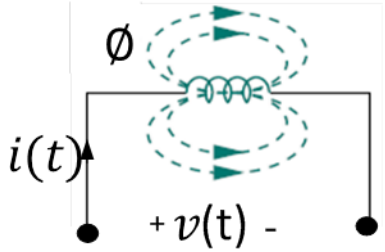
- ✓ Self Inductance
- ✓ Mutual Inductance
- ✓ Dot Convention
- ✓ Energy Stored in a Coupled Circuit
- ✓ Transformers

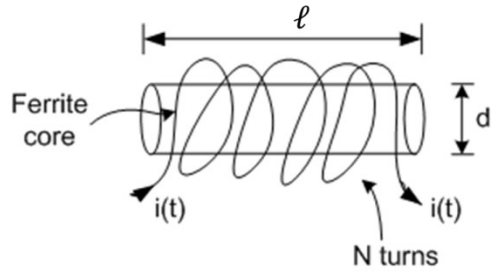
Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Self-Inductance (L)

- ✓ Consider a current $i(t)$ flowing through a coil with N turns. This current produces a magnetic flux $\emptyset\,(weber)$ which is proportional to the current in the coil.





- ✓ The rate at which $N\emptyset$ varies with $i(t)$ is called the self-inductance (L).

$$L = N\frac{d\emptyset}{di}$$

- ✓ Any change in flux induces voltage across the terminals of the coil

$$v(t) = N\frac{d\emptyset}{dt} = N\frac{d\emptyset}{di}\frac{di}{dt} = L\frac{di}{dt}$$

The inductance of the coil is given by

$$L = \frac{N^2\mu A}{\ell}$$

N – number of turns;

μ – the permeability of the core;

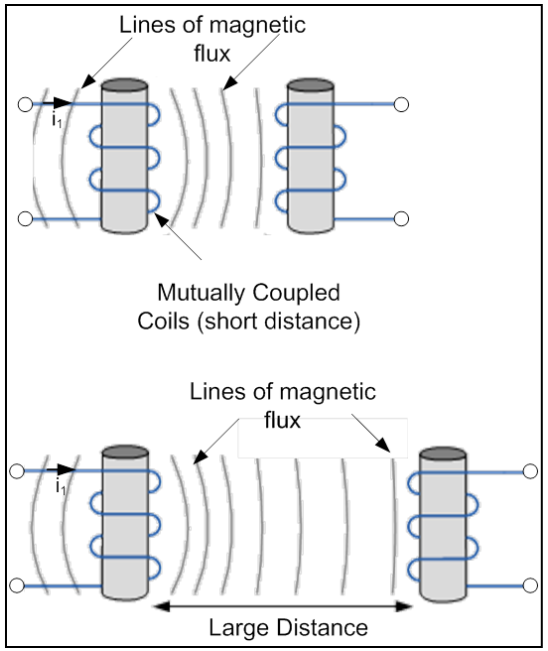
A – the area of the core in m^2 and

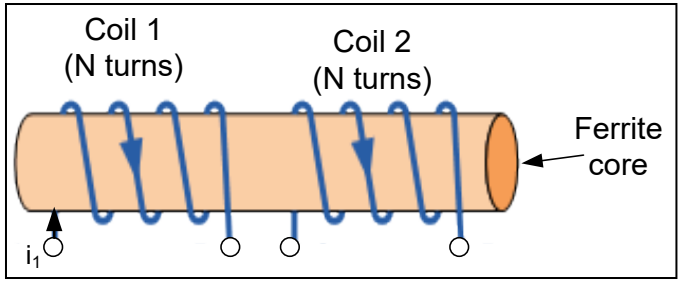
ℓ - the length of the core (m)

Mutual-Inductance (M)

- ✓ A current flowing in one coil establishes a magnetic flux about that coil and also about a second coil nearby.
- ✓ The time-varying flux surrounding the second coil produces a voltage across the terminals of the second coil.
- ✓ This voltage is proportional to the time rate of change of the current flowing through the first coil.

Magnetic coupling through air

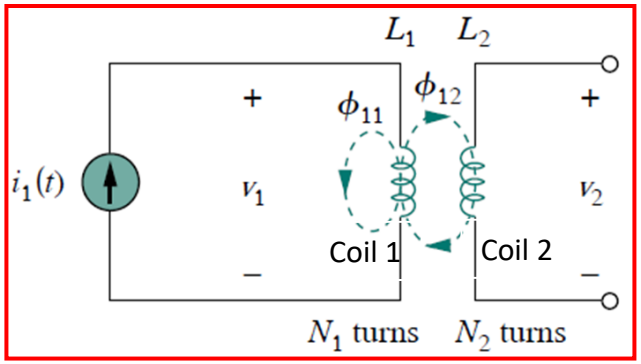




Magnetic coupling through a cylindrical Ferrite core

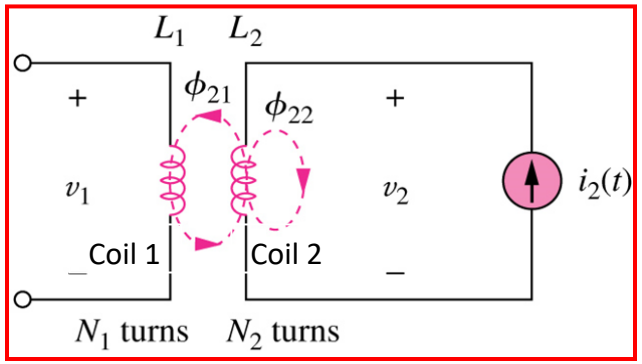
Mutual-Inductance (M)

- ✓ Coil 1 is carrying current i_1, establishes magnetic flux f_1. Some part of this flux, ϕ_{12} links with coil 2 as seen below



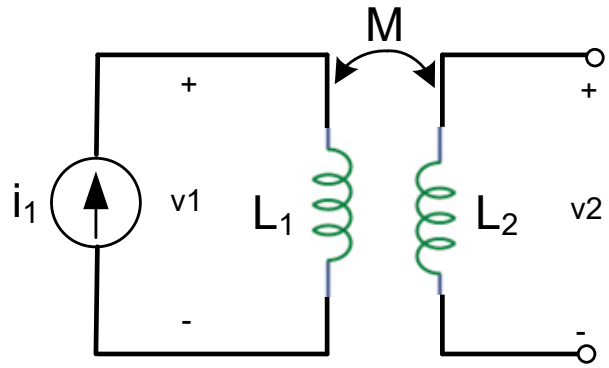
The magnetic flux f_1 has two components - ϕ_{11} which links coil 1 only and ϕ_{12} that links both coils . $f_1 = \phi_{11} + \phi_{12}$

- ✓ Coil 2 is carrying current i_2, establishes magnetic flux f_2. Some part of this flux, ϕ_{21} links with coil 1 as seen below



The magnetic flux f_2 has two components - ϕ_{22} which links coil 2 only and ϕ_{21} that links both coils . $f_2 = \phi_{22} + \phi_{21}$

- ✓ The figure below shows a simple model of two coils L_1 and L_2 sufficiently close together that the flux produced by current $i_1(t)$ flowing through L_1 establishes an open-circuit voltage $v_2(t)$ across the terminals of L_2.



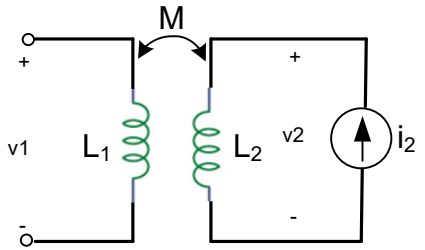
We define mutual inductance M_{12} (=M):

$$v_2(t) = M_{12}\frac{di_1(t)}{dt} \quad \text{and} \quad v_1(t) = L_1\frac{di_1(t)}{dt}$$

- ✓ The order of the subscripts on M_{12}(=M) indicates that a voltage response is produced at L_2 by the current source at L_1 .

- ✓ If the system is reversed as indicated in the diagram below, then we have

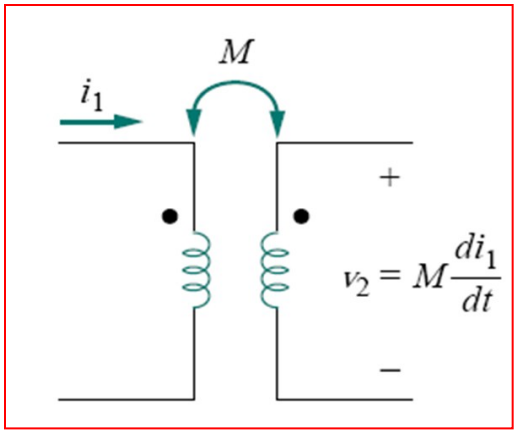
$$v_1(t) = M_{21}\frac{di_2(t)}{dt} \text{ and } v_2(t) = L_2\frac{di_2(t)}{dt}$$

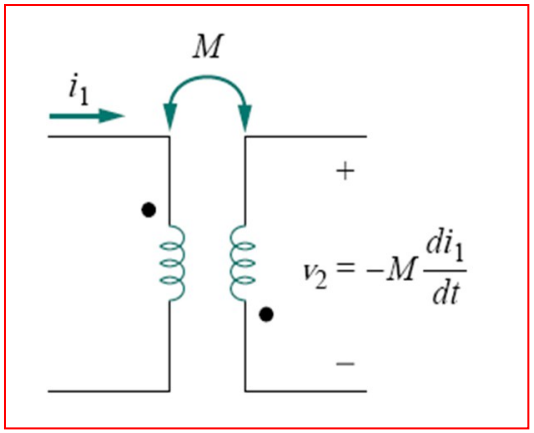


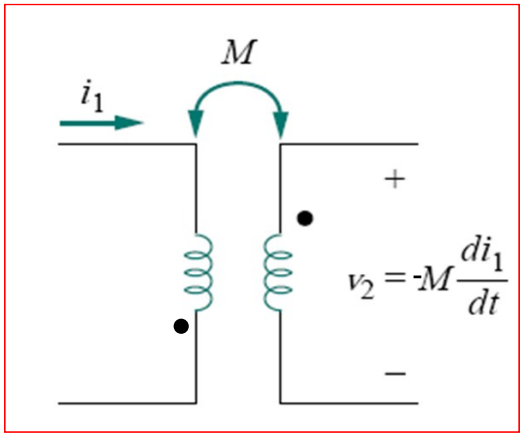
- ✓ It can be shown that the two coefficients of mutual inductance $M_{21} = M_{12} = M$.
- ✓ The Mutual inductance is measured in Henrys (H).
- ✓ An inductor is a two terminal element, whereas the mutual inductance involves four terminals.
- ✓ In order to keep track of the polarities of the mutual voltages, a "dot convention" is normally used.
- ✓ The dot convention makes use of a large dot placed at one end of each of the two coils which are mutually coupled.

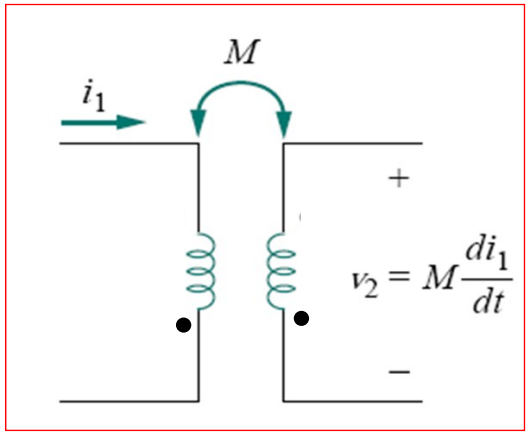
Dot Convention

- ✓ **Dot rule:** When the reference current direction is into the dotted terminal of one coil, the reference polarity of the voltage term induced in the other coil is positive at its dotted terminal.









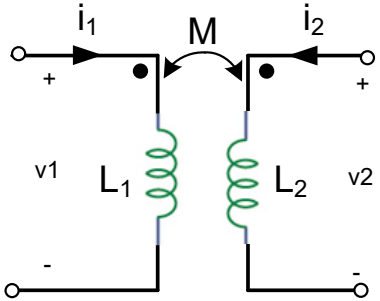
A current entering the undotted terminal of one coil produces a voltage that is positively sensed at the undotted terminal of the second coil.

Combined Mutual and Self-induction Voltage

- ✓ In the portion of the circuit shown below, the currents i_1 and i_2 are shown, each entering the dotted terminal. The voltage (v_1) across L_1 is thus composed of two parts as is the voltage (v_2) across L_2 .

$$v_1(t) = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2(t) = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

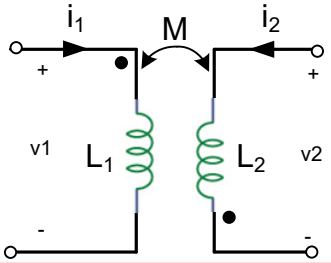


A pair of coupled inductors – both dots at the top

- ✓ In the above circuit, the dots are placed in such a manner that the currents entering (or leaving) both the dotted terminals will produce adding magnetic flux.
- ✓

$$v_1(t) = L_1 \frac{di_1}{dt} + M \frac{d(-i_2)}{dt} = L_1 \frac{di_1}{dt} - M \frac{di_2}{dt}$$

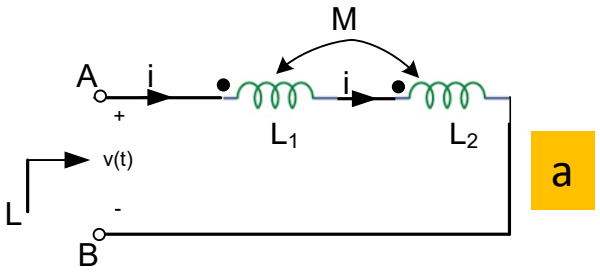
$$v_2(t) = L_2 \frac{di_2}{dt} + M \frac{d(-i_1)}{dt} = L_2 \frac{di_2}{dt} - M \frac{di_1}{dt}$$



One dot on the top and the other at bottom terminal

Example 1

❑ A pair of coupled inductors is connected in two different ways (a) and (b) as shown below. In each case, find the differential equation relating $v(t)$ and $i(t)$ and then find the equivalent inductance seen at terminals A and B.



a

Series-aiding connection

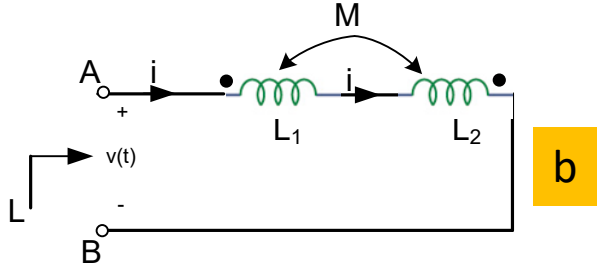
We can directly apply equations obtained in the previous page:

$$v(t) = [L_1 \frac{di}{dt} + M \frac{di}{dt}] + [L_2 \frac{di}{dt} + M \frac{di}{dt}]$$

$$v(t) = (L_1 + L_2 + 2M) \frac{di}{dt}$$

Comparing the relationship:

$$v(t) = L \frac{di}{dt} \text{ ; where } L = L_1 + L_2 + 2M$$



b

Series-opposing connection

We can directly apply equations obtained in the previous page:

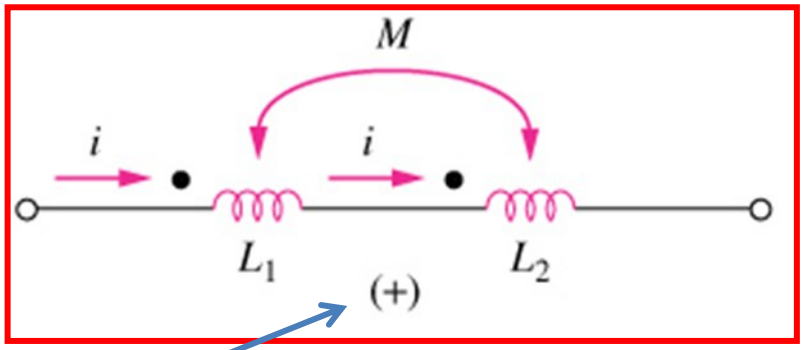
$$v(t) = [L_1 \frac{di}{dt} - M \frac{di}{dt}] + [L_2 \frac{di}{dt} - M \frac{di}{dt}]$$

$$v(t) = (L_1 + L_2 - 2M) \frac{di}{dt}$$

Comparing the relationship:

$$v(t) = L \frac{di}{dt} \text{ ; where } L = L_1 + L_2 - 2M$$

Dot Convention for Coils in Series

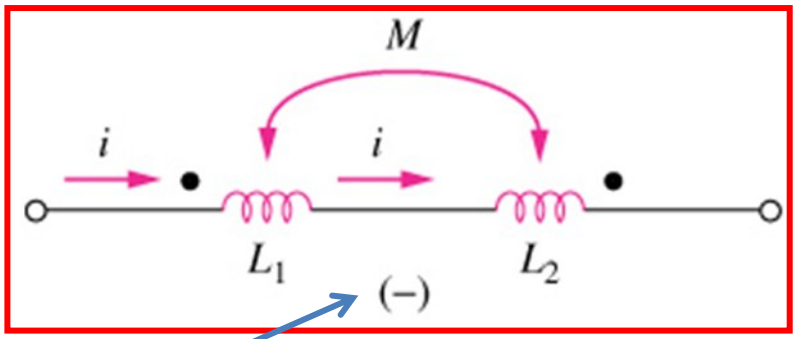


The sign indicates the polarity of the mutual voltage

Series-aiding connection

$$L = L_1 + L_2 + 2M$$

L- Total inductance

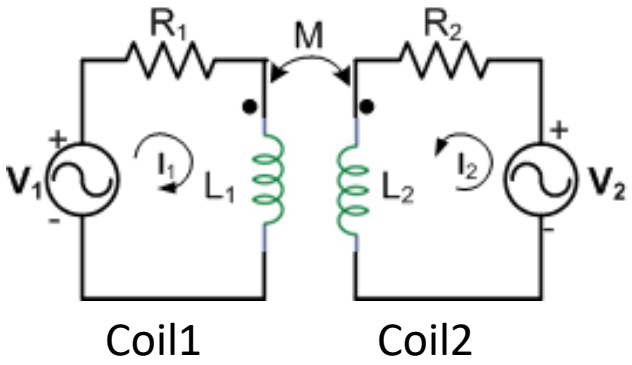


The sign indicates the polarity of the mutual voltage

Series-opposing connection

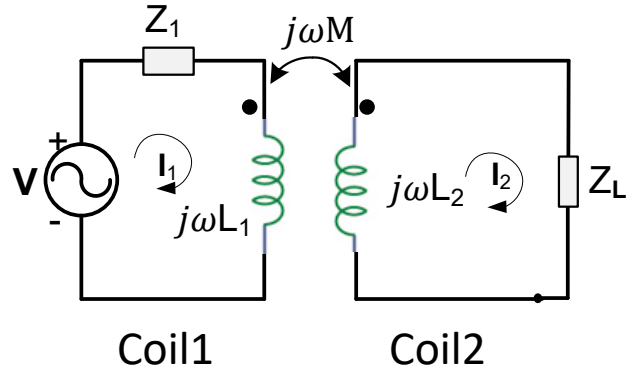
$$L = L_1 + L_2 - 2M$$

Analysis of circuits containing coupled coils



For coil 1: $\boldsymbol{V}_1 = (R_1 + j\omega L_1)\boldsymbol{I}_1 + \mathrm{j}\omega M\boldsymbol{I}_2$

For coil 2: $\boldsymbol{V}_2 = (R_2 + j\omega L_2)\boldsymbol{I}_2 + \mathrm{j}\omega M\boldsymbol{I}_1$



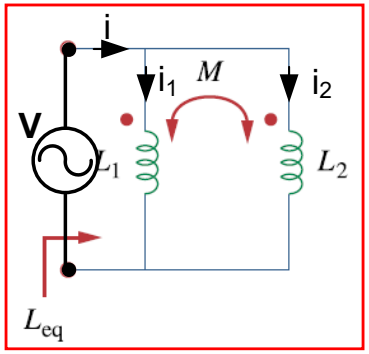
For coil 1: $\boldsymbol{V} = (Z_1 + j\omega L_1)\boldsymbol{I}_1 - \mathrm{j}\omega M\boldsymbol{I}_2$

For coil 2: $0 = (Z_L + j\omega L_2)\boldsymbol{I}_2 - \mathrm{j}\omega M\boldsymbol{I}_1$

Example 2

- For the coupled coils shown below, show that:

$$L_{eq} = \frac{(L_1 L_2 - M^2)}{(L_1 + L_2 - 2M)}$$



Consider the circuit:

$$\boldsymbol{I = I_1 + I_2;} \quad \boldsymbol{Zeq = \frac{V}{I}} \qquad \boldsymbol{\therefore I = \frac{V}{Z_{eq}} = \frac{V}{j\omega L_{eq}}}$$

Applying KVL to each branch gives:

$$\boldsymbol{V} = (j\omega L_1)\boldsymbol{I}_1 + j\omega M \boldsymbol{I}_2$$

$$\boldsymbol{V} = (j\omega L_2)\boldsymbol{I}_2 + j\omega M \boldsymbol{I}_1$$

Solving the above equations for I_1 and I_2 yields

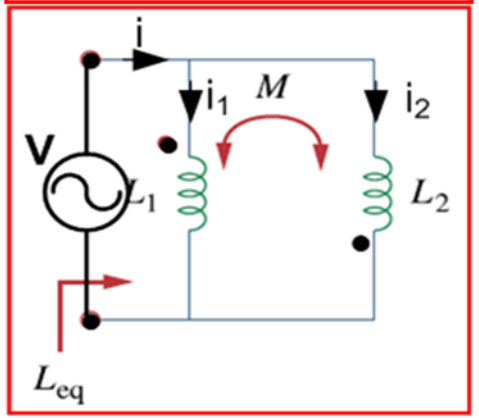
$$\boldsymbol{I_1} = \frac{V(L_2 - M)}{j\omega(L_1 L_2 - M^2)} \qquad \boldsymbol{I_2} = \frac{V(L_1 - M)}{j\omega(L_1 L_2 - M^2)}$$

$$\boldsymbol{I = I_1 + I_2} = \frac{V(L_1 + L_2 - 2M)}{j\omega(L_1 L_2 - M^2)} = \frac{\boldsymbol{V}}{\boldsymbol{j\omega L_{eq}}}$$

$$\therefore L_{eq} = \frac{(L_1 L_2 - M^2)}{(L_1 + L_2 - 2M)}$$

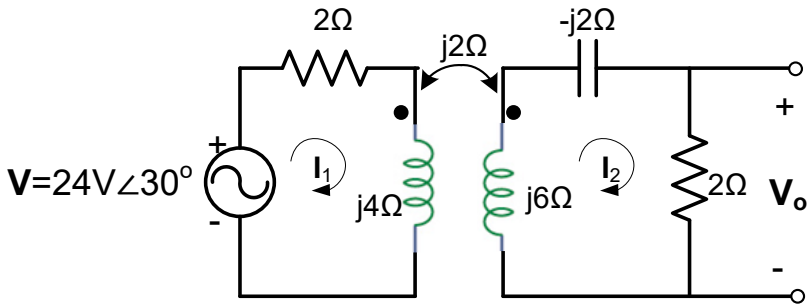
Exercise 1: For the coupled coils circuit, show that:

$$L_{eq} = \frac{(L_1 + L_2 + 2M)}{(L_1 L_2 - M^2)}$$



Example 3

- Determine the output voltage $\mathbf{V}_o$ for the circuit shown below:



Applying KVL equations to the circuit:

$$24\angle 30^0 = (2 + j4)\boldsymbol{I}_1 - \text{j}2\boldsymbol{I}_2$$

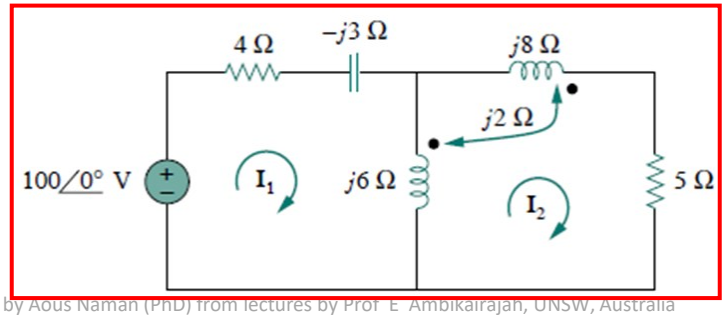
$$0 = (2 + j6 - j2)\boldsymbol{I}_2 - \text{j}2\boldsymbol{I}_1$$

← Solving the these equations for I_2 yields

$$\boldsymbol{I}_2 = 2.68\ A\angle 3.43^0$$

Therefore $V_o = 2\boldsymbol{I}_2 = 5.36\ V\angle 3.43^0$

Exercise 2: Calculate the currents I_1 and I_2 in the circuit shown below:



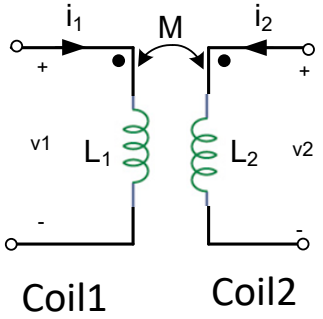
Ans: $\boldsymbol{I}_1 = 20.3\ A\angle 3.5^0$

$\boldsymbol{I}_2 = 8.693\ A\angle 19^0$

Energy in a Coupled Circuit

- ✓ Consider the circuit shown. We assume that the currents i_1 and i_2 are zero initially, so that the energy stored in the coils is zero.
- ✓ If we increase i_1 form zero to I_1 while maintaining i_2 =0, the power in coil 1

$$p_1(t) = \ v_1 i_1 = i_1 L_1 \frac{di_1}{dt}$$



- ✓ The energy stored in the circuit is:

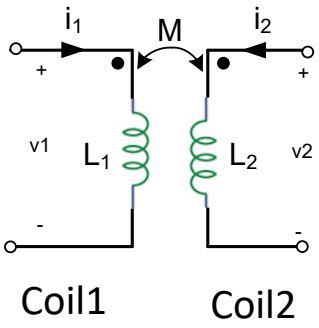
$$e_1 = \int p_1 dt \ = L_1 \int_0^{I_1} i_1 \, di_1 = \frac{1}{2} L_1 I_1^2 \text{ Joule}$$

- ✓ If we now maintain $i_1 = I_1$ and increase i_2 from zero to I_2, the mutual voltage induced in coil1 is $M\frac{di_2}{dt}$ while the mutual voltage induced in coil2 is zero, since i_1 does not change.

- ✓ The power in the coils is now

$$p_2(t) = i_1 v_1 + i_2 v_2 = i_1 M \frac{di_2}{dt} + i_2 L_2 \frac{di_2}{dt}$$

$$p_2(t) = i_1 v_1 + i_2 v_2 = i_1 M \frac{di_2}{dt} + i_2 L_2 \frac{di_2}{dt}$$



✓ The energy stored in the circuit is:

$$e_2 = \int p_2 dt = M I_1 \int_0^{I_2} di_2 + L_2 \int_0^{I_2} i_2 di_2$$

$$e_2 = M I_1 I_2 + \frac{1}{2} L_2 I_2^2$$

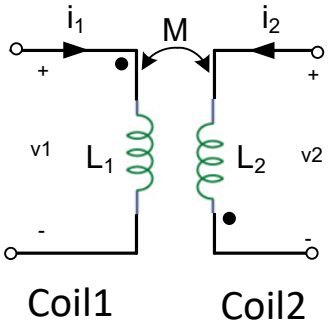
✓ The total energy stored in the coils when both i_1 and i_2 has reached constant values of I_1 and I_2 respectively:

$$e = e_1 + e_2 = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M I_1 I_2$$

← Coil currents both entered the dotted terminals

✓ If one current enters one dotted terminal while the other current leaves the other dotted terminal, the mutual voltage is negative. Therefore

$$e = e_1 + e_2 = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 - M I_1 I_2$$



- ✓ Since I_1 and I_2 are arbitrary values, they may be replaced by i_1 and i_2 which gives the instantaneous energy stored in the circuit

$$e = \frac{1}{2} \mathrm{L}_1 i_1^2 + \frac{1}{2} \mathrm{L}_2 i_2^2 \pm M i_1 i_2$$

- ✓ We will now establish an upper limit for the mutual inductance M.
- ✓ The energy stored in the circuit cannot be negative because the circuit is a passive circuit. Therefore

$$\frac{1}{2} \mathrm{L}_1 i_1^2 + \frac{1}{2} \mathrm{L}_2 i_2^2 - M i_1 i_2 \geq 0$$

- ✓ By adding and subtracting the term $\frac{1}{2} M^2 i_1^2 / \mathrm{L}_2$ the stored energy can be re-written

$$\frac{1}{2}\left[\left(\mathrm{L}_1 - \frac{M^2}{L_2}\right) i_1^2 + \mathrm{L}_2 \left(i_2 - \frac{M}{L2} i_1\right)^2\right] \geq 0$$

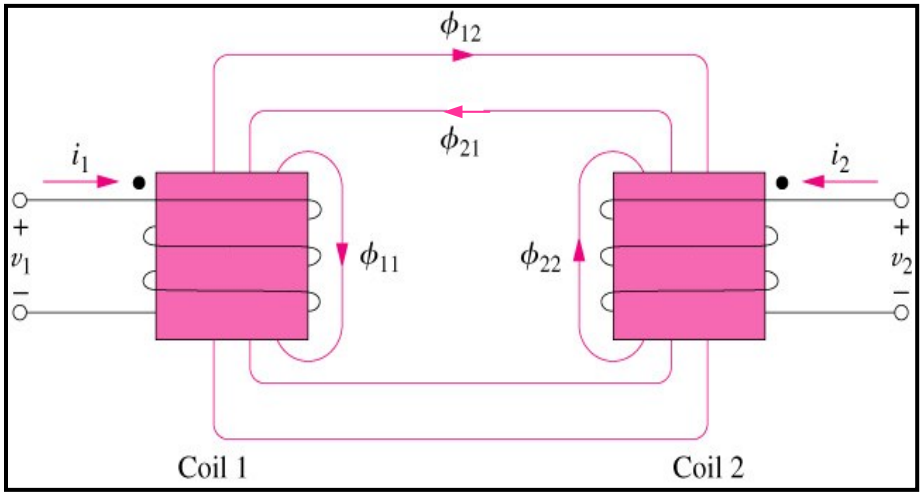
- ✓ To satisfy the above we must require $\mathrm{L}_1 - \frac{M^2}{L_2} \geq 0 \Rightarrow M \leq \sqrt{L_1 L_2}$
- ✓ The mutual inductance cannot be greater than the geometric mean of the self-inductances of the coils

- ✓ The extent to which the mutual inductance *M* approaches the upper limit is specified by the "*coefficient of coupling k*".

$$M = k\sqrt{L_1 L_2} \Rightarrow \quad k = \frac{M}{\sqrt{L_1 L_2}}$$

$$\text{where } 0 \leq k \leq 1 \quad or \quad 0 \leq M \leq \sqrt{L_1 L_2}$$

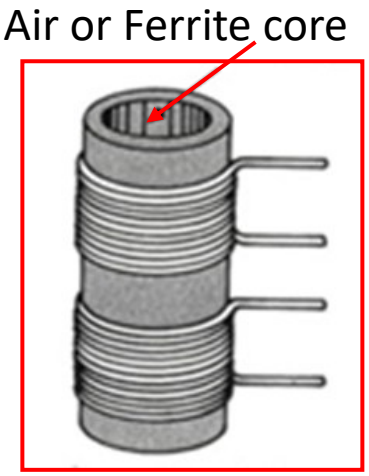
- ✓ The coupling coefficient is the fraction of the total flux emanating from one coil that links the other coil or k is the measure magnetic coupling between two coils (shown below).



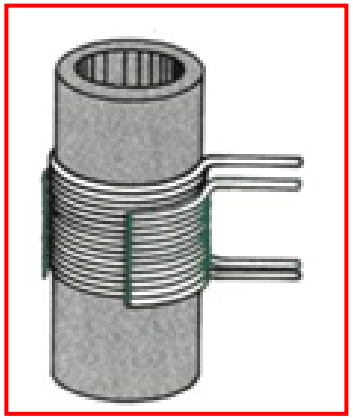
$$k = \frac{\varphi_{12}}{\varphi_{11} + \varphi_{12}}$$

$$k = \frac{\varphi_{21}}{\varphi_{22} + \varphi_{21}}$$

✓ The diagram shows below loosely coupled windings and tightly coupled windings



Loosely coupled windings(k < 0.5)



Tightly coupled windings (k > 0.5)

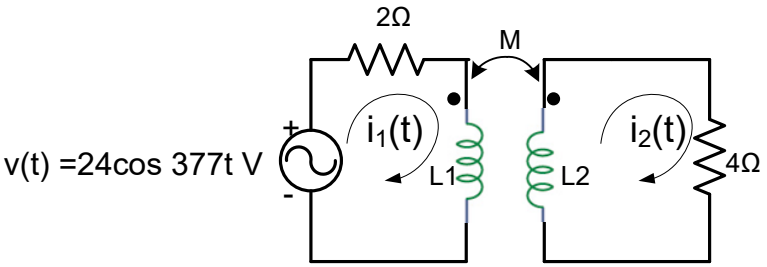
The coupling coefficient *k* depends on :

1. The closeness of the two coils
2. Magnetic properties of the material of their cores
3. Their orientations
4. Their windings arrangements

Note: Perfectly coupled : k = 1

Example 4

- The coupled circuit shown below has a coefficient of coupling of 1 (i.e., k = 1). Determine the energy stored in the mutually coupled inductors at time $t = 5$ ms, given L1 = 2.653 mH and L2 = 10.61 mH.



$j\omega L_1 = j1\boldsymbol{\Omega}$ and $j\omega L_2 = j4\boldsymbol{\Omega}$

$We\ have\ M = k\sqrt{L_1 L_2}$

$M = \sqrt{L_1 L_2}$ = 5.31 mH $(k = 1)$

$j\omega M = j2\boldsymbol{\Omega}$

Mesh equations are: $24\angle 0^0 = (2 + j1)\boldsymbol{I}_1 - \text{j2}\boldsymbol{I}_2$ and $0 = (4 + j4)\boldsymbol{I}_2 - \text{j2}\boldsymbol{I}_1$

By solving the above equations : $\boldsymbol{I}_1 = 9.41 \angle -11.31^0$ A and $\boldsymbol{I}_2 = 3.33 \angle 33.69^0$ A

$\therefore i_1 = 9.41\cos(377t - 11.31^0)$ A and $i_2 = 3.33\cos(377t + 33.69^0)$ A

$at\ t = 5ms$ $i_1 = 9.41\cos(108^0 - 11.31^0) = -1.10$A

$i_2 = 3.33\cos(108^0 + 33.69^0) = -2.61$A

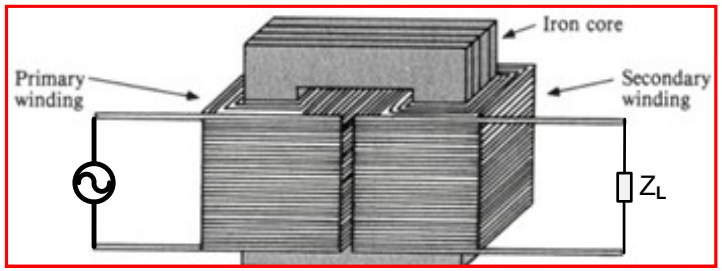
The energy (e) stored in the mutually coupled inductors at t =5ms is:

$$e(t=5) = \frac{1}{2}\text{L}_1 i_1^2 + \frac{1}{2}\text{L}_2 i_2^2 - Mi_1 i_2$$

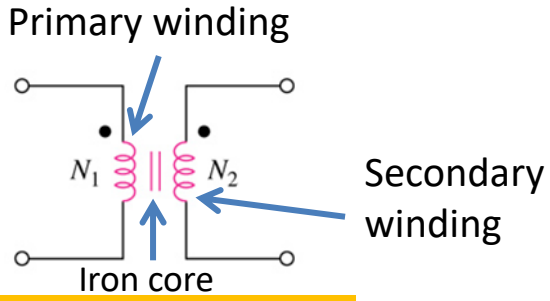
$$= \frac{1}{2}(2.653)(10^{-3})(-1.10)^2 + \frac{1}{2}(10.61)(10^{-3})(-2.61)^2 - (5.31)(10^{-3})(-1.10)(-2.61) = 22.5 \text{ mJ}$$

Transformer

- ✓ A transformer is generally a four-terminal device comprising two (or more) magnetically coupled coils
- ✓ The coil that is directly connected to the input voltage source is called the "primary winding"
- ✓ The coil connected to the load is called the "secondary winding"



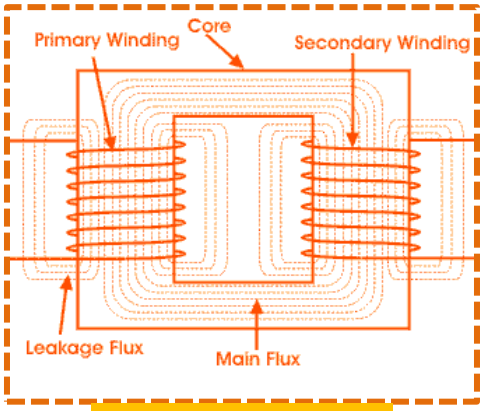
Iron Core Transformer



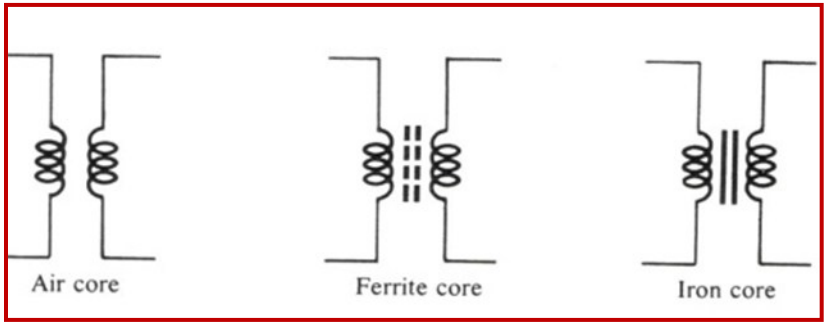
Schematic Symbols

- ✓ Transformers are wound on a common core in several ways
- ✓ The core can be either an air core, an iron core, or a ferrite core
- ✓ Air-core and ferrite core transformers are used for high frequency applications
- ✓ Iron-core transformers are generally used for audio frequency and power applications

- ✓ Iron and ferrite cores increase the coefficient of coupling by providing easy path for magnetic flux
- ✓ In an **ideal transformer**, all of the magnetic flux produced by primary current is assumed to pass through the core to the secondary winding.
- ✓ In practical transformer, some of the magnetic flux lines break out of the core and pass through the surrounding air back to the other end of the windings as illustrated in the diagram below



Flux leakage

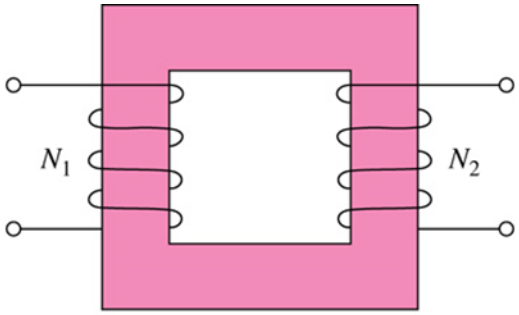


Schematic Symbols

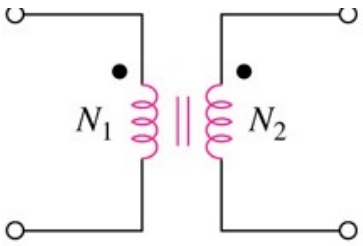
- ✓ Magnetic flux leakage results in reduced secondary voltage
- ✓ Standard schematic symbols for different types of cores

Ideal Transformer

1. Primary and secondary windings have no resistance (loseless).
2. The flux produced by the primary winding is fully coupled to the secondary winding via the core [There is no leakage flux ; hence perfect coupling (i.e. k=1)]
3. Permeability of the core is infinitely large. (coils are wound on highly permeable core)
4. Core Losses are neglected

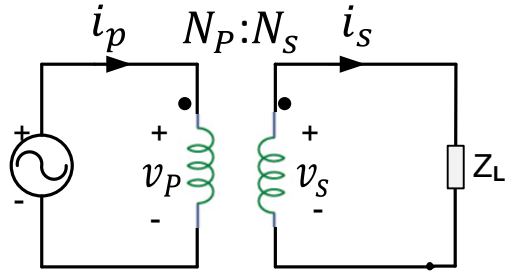


Ideal Transformer



Circuit Symbol

Consider an ideal transformer with the number of turns in the primary winding is (N_P) and the number of turns in the secondary winding is (N_s)



- ✓ $v_P = N_P \frac{d\emptyset}{dt}; \qquad v_S = N_s \frac{d\emptyset}{dt}$
- ✓ $\therefore \frac{v_P}{v_s} = \frac{N_P}{N_S} \quad or \quad \frac{v_s}{v_P} = \frac{N_S}{N_P} = n$
- ✓ n is known as the turns ratio

$$n = \frac{N_s}{N_P}$$

- ✓ For a lossless Transformer, Primary power equals secondary power (i.e. power must be conserved)

$$v_P i_P = v_s i_s \Rightarrow \frac{i_s}{i_p} = \frac{v_P}{v_s} = \frac{N_P}{N_S} = \frac{1}{n}$$

$\frac{v_P}{v_s} = \frac{N_P}{N_S} = \frac{1}{n}$

$\frac{i_P}{i_s} = \frac{N_S}{N_p} = n$

N_P:N_s = 1 : n

Step-up Transformer

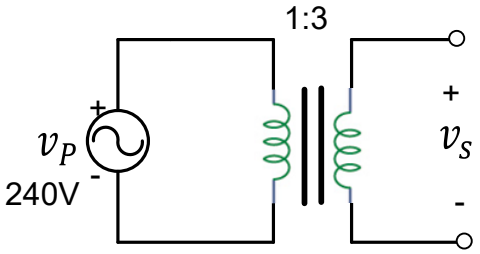
- ✓ A step-up transformer is used to increase the ac voltage. A transformer in which the secondary voltage is greater than the primary voltage is called a step-up transformer.

$$\frac{v_S}{v_P} = \frac{N_S}{N_P} \Rightarrow v_S = \left(\frac{N_S}{N_P}\right) v_P \Rightarrow \quad v_S = n\ v_P$$

- ✓ Secondary voltage is equal to the turns ratio (*n*) times the primary voltage (assuming the coefficient of coupling (k) = 1)
- ✓ Since the number of turns in the secondary winding is always greater than the number of turns in the primary winding, the turns ratio (*n*) for a step-up transformer is always > 1

Example 5

- ❑ The transformer circuit shown below has 400 turns in the primary winding and 1200 turns in the secondary winding. What is the voltage across the secondary winding?



$$n = \frac{N_S}{N_P} = \frac{1200}{400} = 3 \qquad v_S = n\ v_P$$

$$v_S = 3\ (240) = 720\text{V}$$

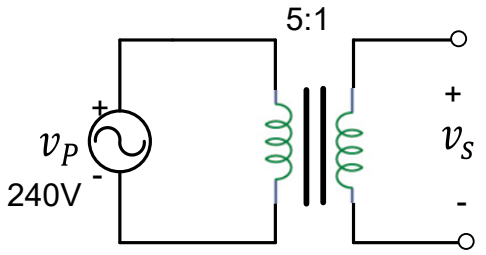
Note: The turns ratio of 3 is indicated on the schematic as 1:3, meaning that there are 3 secondary turns for each primary turn.

Step-down Transformer

- ✓ A step-down Transformer is used to decrease ac voltage. A transformer in which the secondary voltage is less than the primary voltage is called a step-down transformer.
- ✓ Since the number of turns in the secondary winding is always less than the number of turns in the primary winding, the turns ratio (n) for a step-up transformer is always < 1.
- ✓ Note: Since coupling between the input and the output ports of a transformer is exclusively magnetic, transformers are also used to provide electrical isolation.
- ✓ Transformers with n=1 are used exclusively for this purpose and are called **isolation transformers**.

Example 6

❑ The transformer circuit shown below has 100 turns in the primary winding and 20 turns in the secondary winding. What is the voltage across the secondary winding?



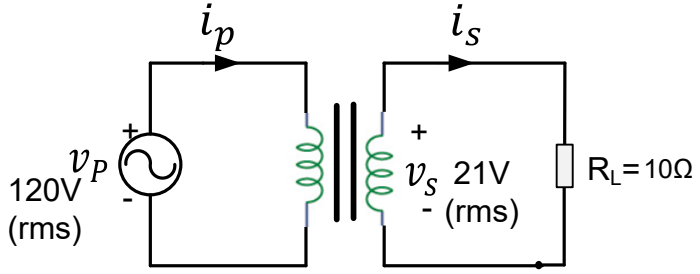
$$n = \frac{N_S}{N_P} = \frac{20}{100} = 0.2 \quad v_s = n\ v_P$$

$$v_s = 0.2\ (240) = 48\text{V}$$

Note:
n > 1 step-up transformer ($v_s > v_P$)
n < 1 step-down transformer($v_s < v_P$)

Example 7

- A transformer is to be used to step down the 120V(rms) utility voltage to an ac voltage of 21V (rms) for a 10 ohm load. Find turns ratio (*n*) and the rms values of the currents in the two windings.



For an ideal transformer :

$$\frac{v_s}{v_P} = \frac{N_s}{N_P} = \frac{i_p}{i_s}; \quad n = \frac{v_s}{v_P}$$

$$n = \frac{v_s}{v_P} = \frac{21}{120} = 0.175$$

$$i_s = \frac{v_s}{R_L} = \frac{21V}{10} = 2.1A\ (rms)$$

$$i_p = \left(\frac{N_s}{N_P}\right) i_s = (0.175)(2.1) = 0.367A\ (rms)$$

Example 8

- An ideal transformer is rated at 4800/240 V, 9.6 kVA, and has 100 turns in the secondary winding. Calculate (i) the turns ratio (ii) the number of turns in the primary winding (iii) the primary and secondary currents.

This is a step down transformer since v_P = 4800 V and v_s = 240 V

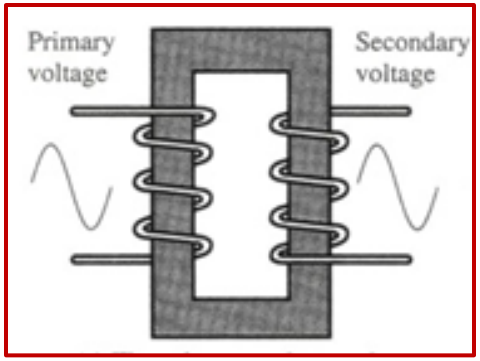
$$n = \frac{v_s}{v_P} = \frac{240}{4800} = 0.05 \quad and \quad n = \frac{N_s}{N_P} \Rightarrow N_P = \frac{100}{0.05} = 2000\ turns$$

For an ideal transformer power must be conserved: $v_P i_P = v_s i_s = 9.6kVA$

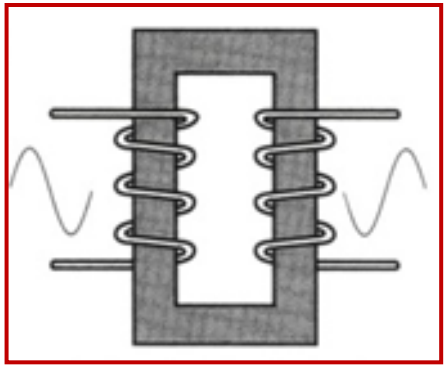
$$i_p = \frac{9600}{4800} = 2A; i_s = \frac{9600}{240} = 40A \quad or \quad \frac{N_s}{N_P} = \frac{i_p}{i_s} \Rightarrow i_s = \frac{i_p}{n} = \frac{2}{0.05} = 40A$$

Direction of windings

- ✓ Another important transformer parameter is the direction in which the windings are placed around the core.
- ✓ The direction of the windings determines the polarity of the secondary voltage with respect to the primary voltage.



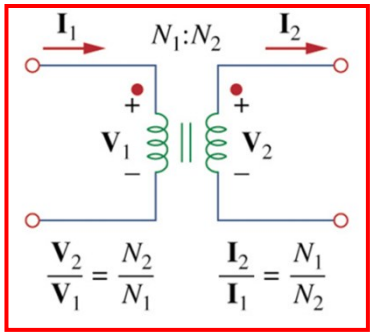
Primary and secondary voltages are in phase when the windings are in the same effective direction



Primary and secondary voltages are 180^0 out of phase when the windings are in the opposite direction

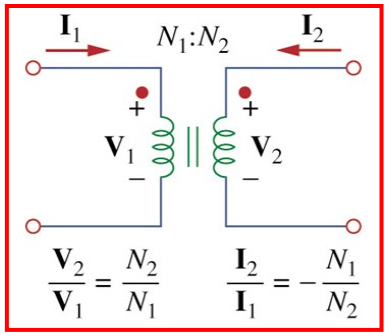
Dot Rule for transformers

- ✓ It is important that we know how to get the proper polarity of the voltages and the direction of the currents for a transformer
- ✓ If the dots are omitted, it implies that the phase relationship between the two primary and the secondary windings are unimportant.



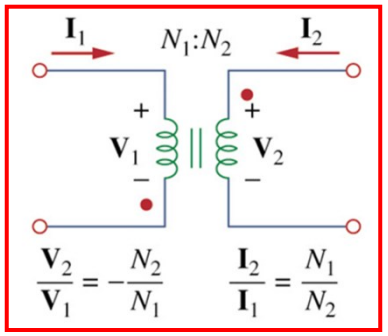
V_1 and V_2 are both positive at the dotted terminals.

I_1 enters and I_2 leaves the dotted terminal.



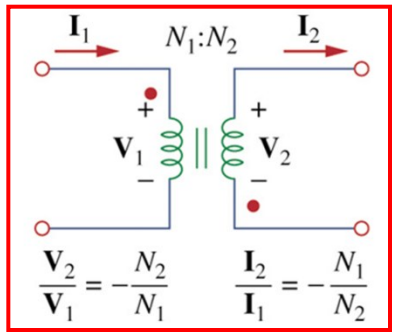
V_1 and V_2 are both positive at the dotted terminals.

Both I_1 and I_2 enter the dotted terminals.



V_1 is negative and V_2 is positive at the dotted terminal.

I_1 leaves and I_2 enters the dotted terminal.



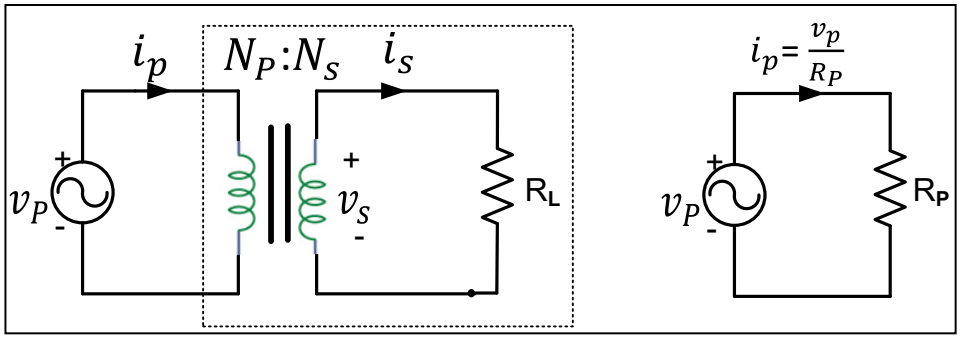
V_1 is positive and V_2 is negative at the dotted terminal.

Both I_1 and I_2 enter the dotted terminals.

- ✓ If I_1 and I_2 both enter into or both leave the dotted terminals, use − for turns ratio. Otherwise, use +.
- ✓ If V_1 and V_2 are both positive or both negative at the dotted terminals, use + for turns ratio. Otherwise, use −.

Reflected Load

- ✓ The concept of *reflected load* is illustrated in the diagram below.
- ✓ The load (R_L) in the secondary circuit of a transformer is reflected into the primary circuit by transformer action.
- ✓ The primary circuit sees a resistance (R_P) determined by the turns ratio and the actual value of the load resistance (R_L). R_P is now called the reflected resistance.



Actual Load

Reflected Load(R_P)

The resistance in the primary circuit is: $R_P = \frac{v_P}{i_P}$

The resistance in the secondary circuit is: $R_L = \frac{v_s}{i_s}$

$$\therefore \frac{R_P}{R_L} = \frac{\frac{v_P}{i_P}}{\frac{v_S}{i_S}} = \frac{v_P}{v_s}\frac{i_s}{i_p} = \frac{N_P}{N_s}\frac{N_P}{N_s} = \left(\frac{N_P}{N_s}\right)^2$$

$$\therefore R_P = R_L \left(\frac{N_P}{N_s}\right)^2 = \frac{R_L}{n^2}$$

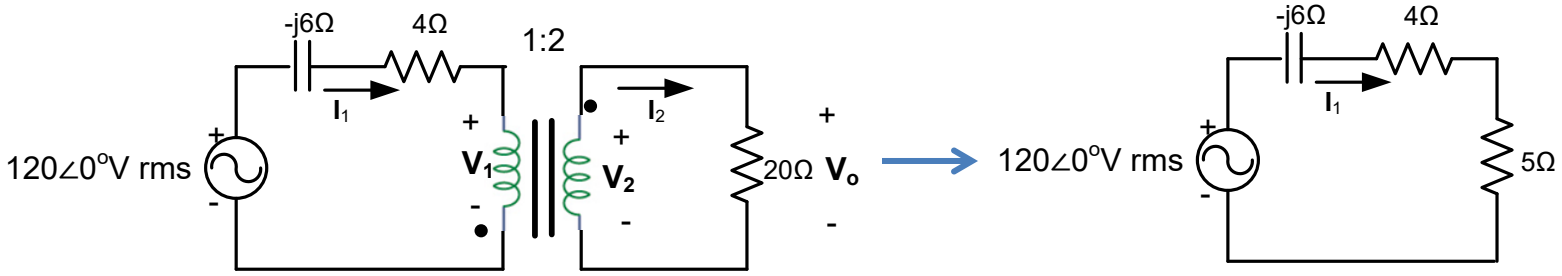
Reflected resistance

Note: If the load is an impedance (Z_L) then the reflected impedance (Z_P) is given by:

Reflected impedance → $Z_P = \frac{Z_L}{n^2}$

Example 9

- For the ideal transformer circuit shown below Find (a) the source current $\mathbf{I}_1$, (b) the output voltage $\mathbf{V}_o$ and (c) the complex power supplied by the source.



$$Reflected\ impedance\ Z_R = \frac{20}{n^2} = \frac{20}{2^2} = 5\Omega$$

$$Primary\ circuit\ impedance\ Z_{in} = 4 - j6 + 5\Omega = 9 - j6 = 10.82\angle -33.690\ \Omega$$

$$I_1 = \frac{V}{Z_{in}} = \frac{120\angle 0^o}{10.82\angle -33.69^0} = 11.09\angle 33.69^o\ \text{A}$$

- Both I_1 and I_2 leave the dotted terminals, use − for turns ratio

$$I_2 = -\frac{I_1}{n} = -5.545\angle 33.69^o\ \text{A} \qquad V_0 = 20\boldsymbol{I}_2 = 110.9\angle 213.69^o\ \text{V}$$

$$The\ complex\ power\ Supplied:$$
$$\boldsymbol{S} = Vs\boldsymbol{I_1}^* = (120\angle 0^o)\ (11.09\angle -33.69^o) = 1330.8\angle -33.69^o\ \text{VA}$$

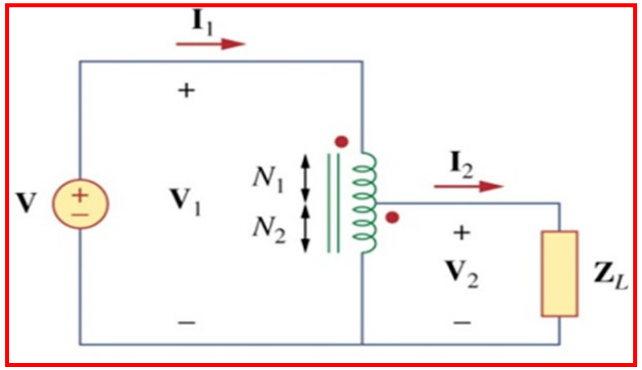
Transformer Rating and efficiency

- ✓ A transformer is typically rated in volt-amperes (VA), primary/secondary voltage and operating frequency
- ✓ For example, a given transformer rating may be specified as 2 kVA, 1000/100, 50Hz.
- ✓ For example that 100 V is the secondary voltage, then the load current (i_L) is
- ✓ $i_L = \frac{P_S}{v_S} = \frac{2\ kVA}{100V} = 20A$ This is the maximum current that the secondary can handle.

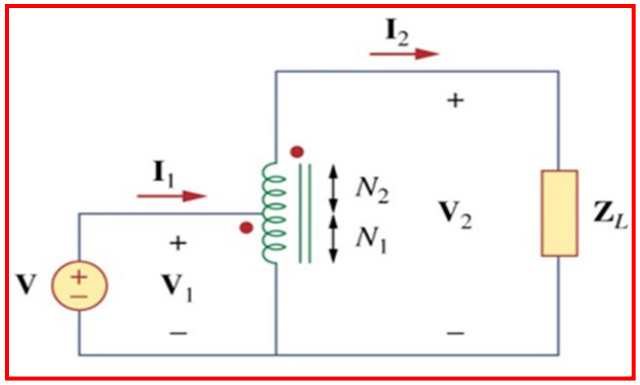
- ✓ In an ideal transformer, the secondary power is equal to the primary power
- ✓ Because of the non-ideal characteristics of the transformer, always there will be power loss.
- ✓ The secondary (output) power is always less than the primary (input) power
- ✓ The efficiency (η) of a transformer is a measure of the percentage of the input power that is delivered to the output.
- ✓ $\eta = (\frac{P_{out}}{P_{in}})100\%$

Auto Transformer

An auto transformer is a transformer in which both the primary and the secondary are in a single winding



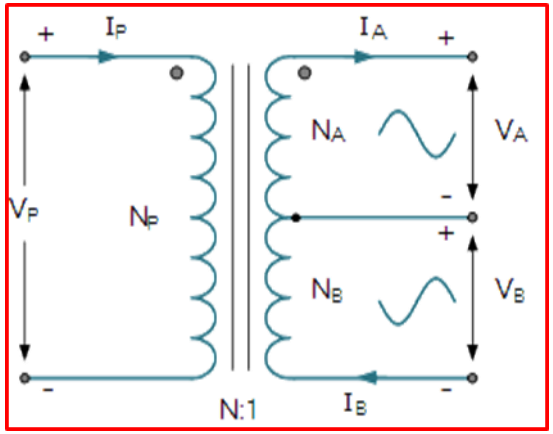
$$\frac{\mathbf{V_1}}{\mathbf{V_2}} = \frac{N_1 + N_2}{N_2}$$

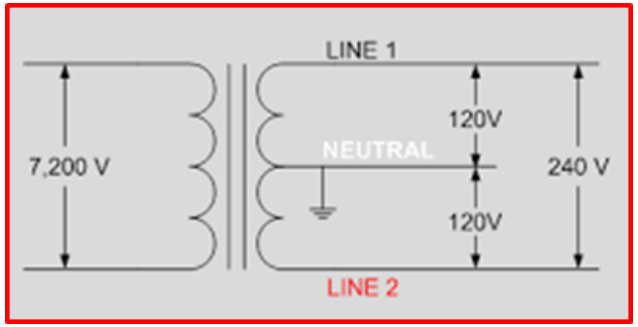


$$\frac{\mathbf{V_1}}{\mathbf{V_2}} = \frac{N_1}{N_1 + N_2}$$

Tapped Transformer

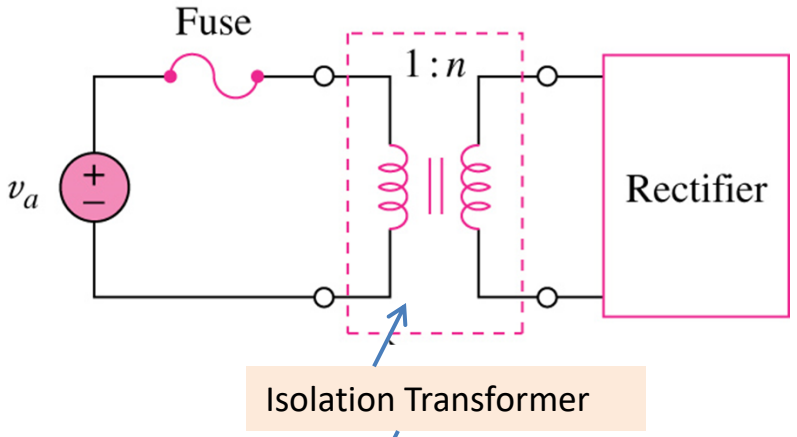
Transformers with a centre-tapped secondary winding is shown below



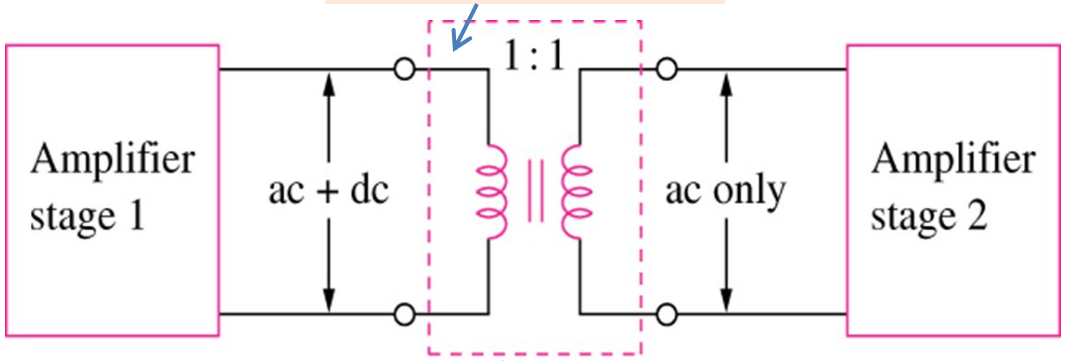


Transformer Applications

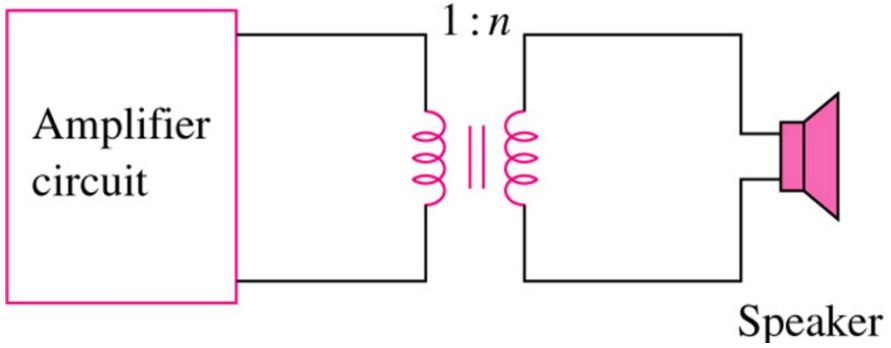
- ✓ Transformer as an Isolation device to isolate ac supply from a rectifier



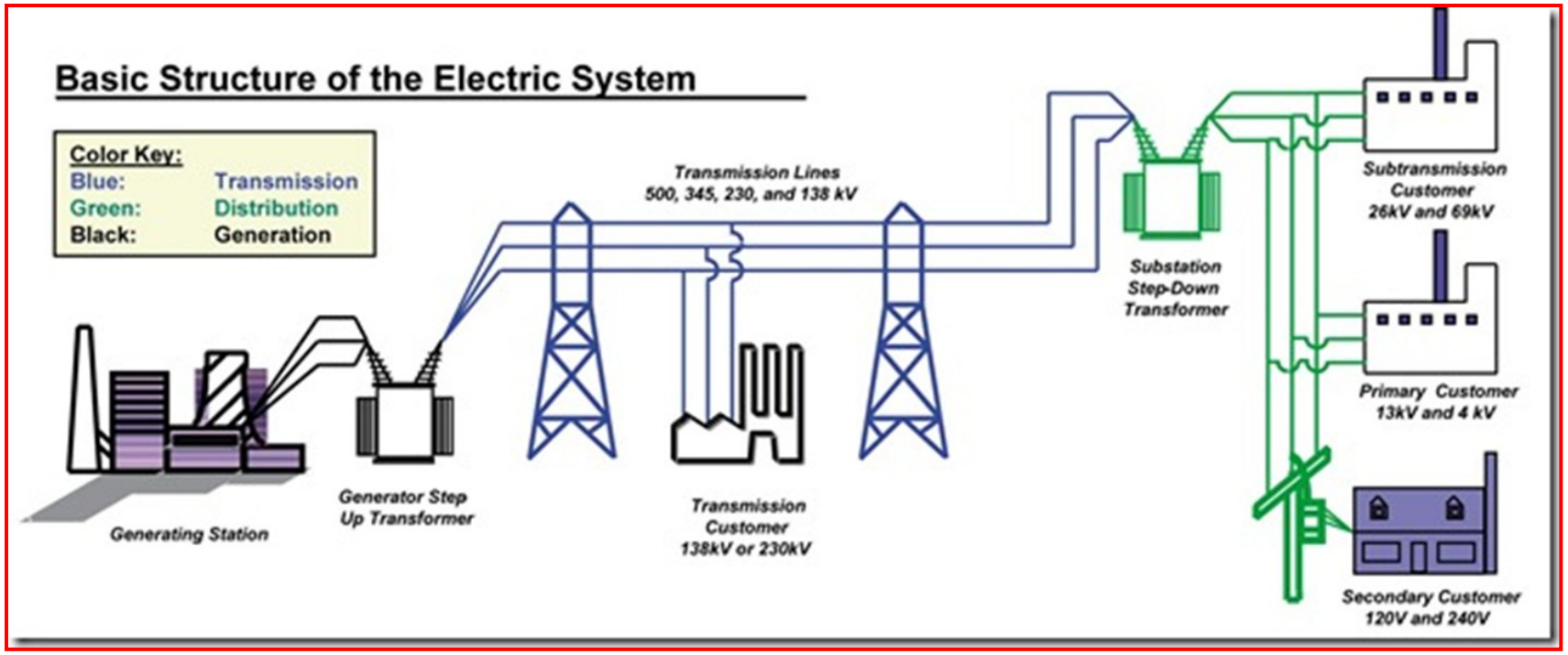
- ✓ Transformer as an Isolation Device to isolate dc between two amplifier stages.

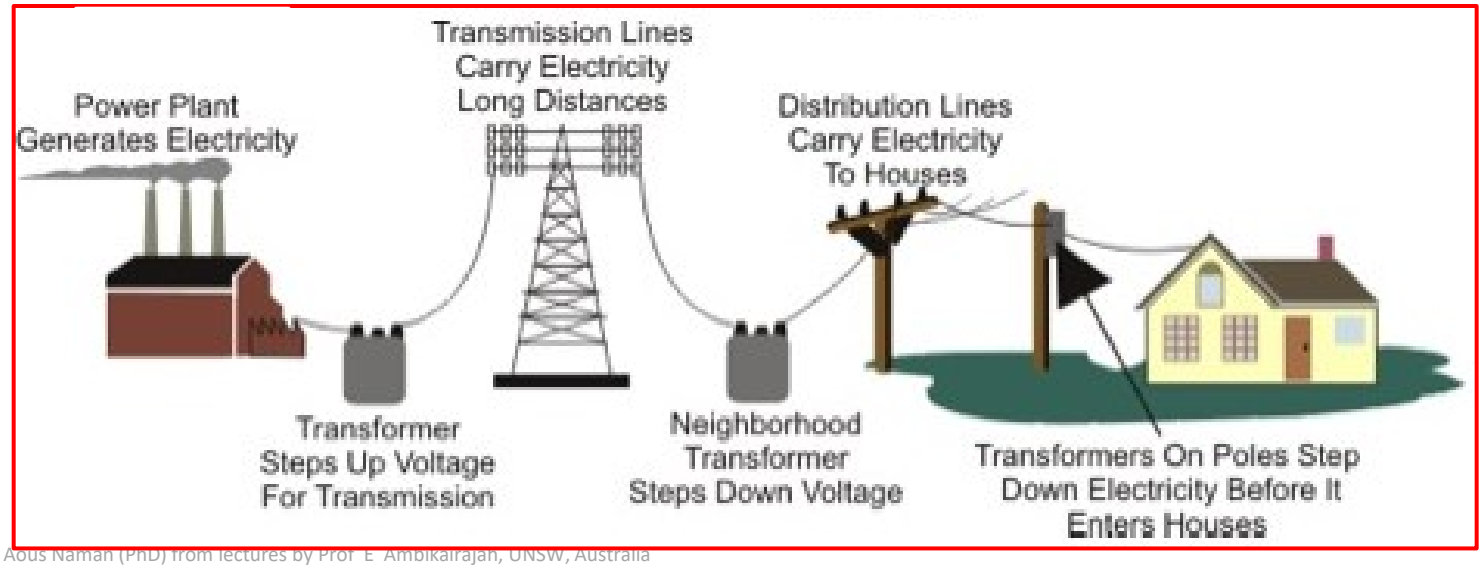


- ✓ Transformer coupling for maximum power transfer



Power distribution System





ELEC2134 : Circuits and Signals (PART B)

Transformers and coupled coils
Tutorial Questions

Magnetically Coupled Circuits

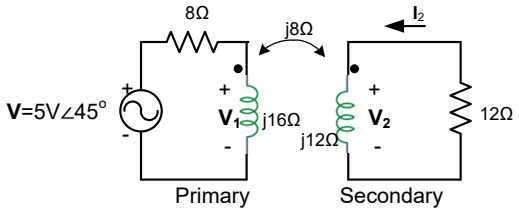
- ✓ Self Inductance
- ✓ Mutual Inductance
- ✓ Dot Convention
- ✓ Energy Stored in a Coupled Circuit
- ✓ Transformers

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

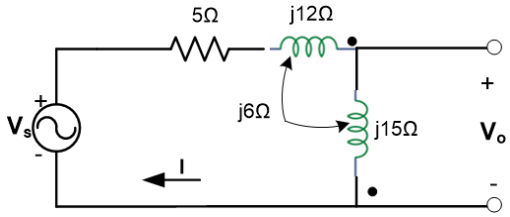
Tutorial Questions (Transformers and Coupled Coils)

Question 1 (Pass-Level): Find the voltage $v_2(t)$ in the circuit shown below:



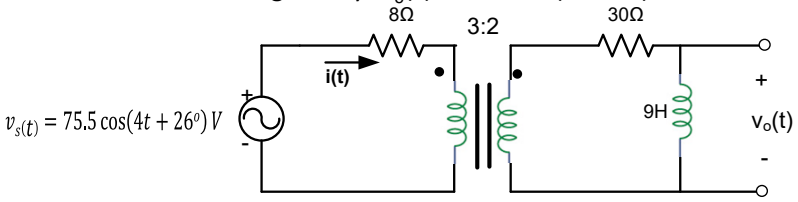
Ans: $v_2(t) = 1.656\cos(4t+39^o)$ V

Question 2 (Pass-Level): Determine the output voltage $v_0(t)$ for the circuit shown below, where the input $v_s(t) = 5.94\cos(3t + 140^o)\ V$

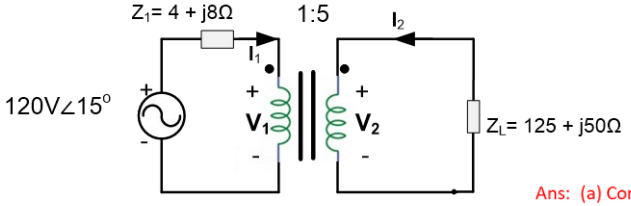


Ans: $v_o(t) = 3.17\cos(3t+147^o)$ V

Question 3 (Pass-Level): The input voltage for the circuit below is given by: $vs(t) = 75.5\cos(4t + 26^o)\ V$. Show that the output voltage across the 9H inductor is given by: $v_o(t)$=36.82 cos(4t+69°)V .

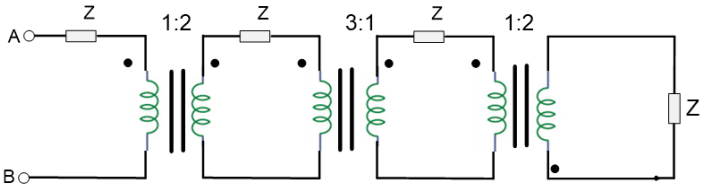


Question 4 (Pass-Level): Determine the value of the complex power (a) supplied by the voltage source (b) received by the load Z_L and (c) received by the transformer in the circuit shown below.



Ans: (a) Complex power delivered= 358+j398 VA
(b) -198.9 –j79.56 (c) 0 VA

Question 5 (Pass-Level): Determine the impedance Z_{AB} for the circuit given below.



Ans: $Z_{AB} = 4.063Z$